Class C Dx Cost Estimator: Modeling Pipeline Architecture

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1. Purpose and scope

This pipeline trains, validates and blends models that predict `adjusted_net_total` for the Class C Dx Estimator, using only the 20 Power App inputs. It runs on Abul's side because the data cannot be shared; the code is built to be scheduled in the background and to leave a comparable log on every run.

- **In scope:** leakage-safe features, frozen validation, a diverse model zoo, hill-climb blending, calibration, input ablation, and a reproducible results log.
- **Out of scope:** deploying to the Power App, retraining governance, and changing the business acceptance rule.

2. Business KPI and scoring rule

Success means at least 68% of estimates land within ±50% of the eventual Class A customer-payable amount, per the agreed-accuracy draft. The pipeline scores every model, blend and fold with this one function so results are comparable.

```
APPE = \frac{|Prediction - ClassA|}{ClassA}, \quad hit = (APPE \le 0.5)
```

Rule	Definition
Primary KPI	Share of records with APPE ⩽ 0.5; target ⩾ 68%
Zero actuals	APPE is undefined, so a record is a hit when absolute dollar error is under $500
Secondary KPI	Share of positive records predicted more than 10% under actual (lower is better)
Not the target	The earlier −10% / +50% window at 70% was an experimental, stricter objective

The evaluation counts records, not dollars, so a $1.48M job weighs the same as a $2K job. That is why every model trains on log1p of the target.

3. Data and input contract

The training file has 10,499 rows and 49 columns; SEC is 9,738 rows, LE 437 and PRI 324. The model may read only the 20 Power App inputs below, plus features derived from them.

Group	Fields
Continuous	`AWF`, `exp_length_oh`, `exp_length_ug`, `exp_length_su`
Counts	`num_of_work_locations`, `number_of_poles_installed`, `number_of_poles_removed`
Binary flags	`is_overhead`, `is_underground`, `far_or_same_side`, `new_or_existing_transformer`, `rate_category_group_Residential`
One-hot: sub-work-type	`LE`, `PRI`, `SEC`
One-hot: amps group	Less Than 200A, 200A - 400A, Greater Than 400A
One-hot: phase	`Single`, `Three`

Always excluded: `project_cost`, `project_cost_cleaned`, `investment_dollars`, `investment_dollars_cleaned` (the target is their difference), `work_order_number` (ID), `QMART` / `QMCOD` / `QMGRP`, `is_street_light` and `less_than_2_cu` (constant), and the full `rate_category`.

Zero targets: SEC 17.7%, PRI 6.5%, LE 0%. The target is therefore a mixture, which drives the hurdle models in section 6.

4. Leakage controls and validation

Every score must come from a split frozen on disk, so a gain between two runs is a real gain and not a new random draw.

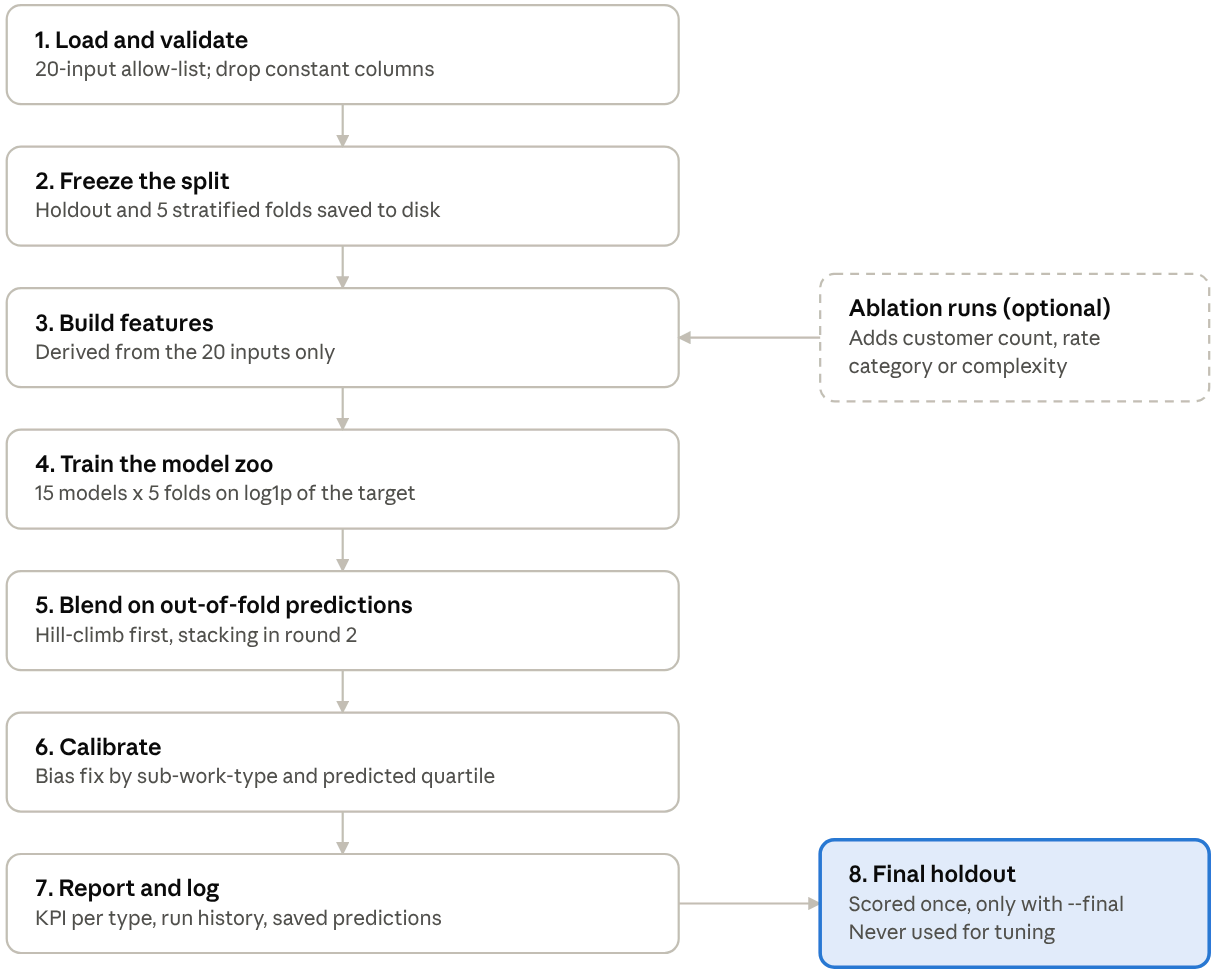
1. **Feature allow-list.** The script reads only the 20 inputs and stops with a clear message if a column is missing. Derived features use inputs only (totals, shares, ratios, poles net, AWF × length).
2. **Frozen holdout.** A fixed share of rows is set aside once and saved to `outputs/split.json`. It is scored only with `--final`, once, at the end.
3. **Frozen stratified CV.** Five folds, stratified by sub-work-type × target bucket (zero plus quartiles of positive cost), saved in the same file.
4. **Out-of-fold only.** Blend weights, calibration and any target encoding are fit on out-of-fold predictions, never on the training fit.
5. **Alignment with the existing work.** The current Copilot model uses a 30% holdout; this pipeline defaults to 15%. We should reuse the existing holdout (by work order) so scores compare directly.

Time-based option: the file has no date column. If `work_order_number` increases with time, it can serve as a proxy for a train-on-older, test-on-newer split, which is the honest test for a production estimator. This needs confirming first.

5. Pipeline architecture

Every tuning decision is made on out-of-fold predictions, and the holdout is scored once, last.

Every decision uses out-of-fold data; the holdout is scored once



pipeline · 8 stages, 1 optional branch

Stages 1 to 7 run on every scheduled run. The ablation branch feeds extra input variants into stage 3, and stage 8 runs only when you pass `--final`.

6. Model zoo

The pipeline trains 15 models chosen to make different kinds of errors. All except Tweedie and the hurdle models train on log1p of the target; every model outputs log predictions so any subset can be blended.

Family	Models	Why it is included
Boosting	`lgbm`, `xgb`, `cat`, `hgb`	Strong tabular baselines

Family	Models	Why it is included
Boosting variants	`lgbm_huber`, `cat_mae`	Robust to outliers; MAE seeks the median, which suits a hit-rate KPI
Bagged trees	`et`, `rf`	Different splitting and variance behaviour
Zero-aware	`xgb_tweedie`, `hurdle_soft`, `hurdle_hard`	Handle the $0 mixture directly
Linear	`ridge_poly`	Smooth degree-2 interactions, very different bias
Kernel, distance, neural	`svr`, `knn`, `mlp`	Similarity-based and smooth views the trees cannot give

Hurdle models. A classifier estimates P(cost > 0) and a regressor, trained on positives only, estimates the amount. `hurdle_hard` returns $0 when P < 0.5, which helps hits on positives; `hurdle_soft` returns the expected value.

Component models (planned). Separate models for project cost and investment dollars, with their difference added to the blend as an extra member. The difference is not used as the primary estimate, because subtracting two noisy predictions is least reliable near $0, where SEC has about 17% of its rows.

Sub-work-type specialists (planned). LE and PRI have few rows, so specialists are blended in by out-of-fold evidence rather than replacing the shared models.

7. Ensembling and calibration

The blend is chosen on out-of-fold predictions only, so it cannot see the holdout.

1. **Hill-climb.** Greedy forward selection with replacement, averaging in log space. Each step adds the model that most improves the score: hit rate minus 0.25 × the share more than 10% under.
2. **Stacking (round 2).** A small meta-learner on the out-of-fold predictions, compared with hill-climb under repeated CV.
3. **Calibration (round 2).** A multiplicative bias correction by sub-work-type × predicted-cost quartile, fit out-of-fold. It targets the secondary KPI: fewer severe under-estimates without losing hits.

4. **Multi-seed averaging (round 2).** Repeat CV with several seeds so a gain of half a point is separated from noise.

The score is a hit rate over a few thousand records, so differences under about one point should be treated as noise until repeated CV confirms them.

8. Ablation: how much do the missing inputs matter?

The strongest cost drivers in the data are not in the 20-input contract. The ablation mode quantifies the lift each one would give, so the business can decide whether to capture it in the Power App.

Run	Inputs	Question it answers
A0	The 20 inputs	Current achievable hit rate
A1	A0 + `number_of_cus`	Value of customer count (0.86 correlation with project cost)
A2	A0 + full `rate_category`	Value of the full rate category (medians differ up to 12×)
A3	A0 + complexity category	Value of complexity
A4	A0 + all three	Upper bound

Only fields confirmed to be known at estimate time may become production inputs. The ablation result is evidence for a business request, not permission to use the field.

9. Background runner

The final script is a single file that can run unattended and leaves a comparable record of every run.

- **Config block** at the top: data path, time budget, models to run, number of seeds, ablation on or off.
- **Time budget:** models run cheapest first; when the budget is reached the run stops cleanly, blends what finished, and logs what was skipped.
- **Logging:** a timestamped log file per run, plus one row appended to `outputs/run_history.csv` with git-style run id, models, seeds, OOF hit rate, under-10% rate and per-type scores.

- **Artifacts:** `split.json`, `oof_predictions.csv`, `blend_weights.json`, and a per-sub-work-type report.
- **Fail-safe:** missing columns, constant columns and non-numeric values are reported up front, and one failing model never stops the others.

Open item: where it runs (Azure ML, Fabric, a VM or a laptop) decides CPU cores and which libraries can be installed, and so the time budget.

10. Risks, open questions and roadmap

The biggest risk is a label mismatch: if the model trains on cleaned values but Class A comes from raw SAP, scores will not transfer.

Risk or question	Why it matters	Who answers
Does Class A equal `adjusted_net_total`, or raw or cleaned cost?	Cleaning changed project cost on 53.7% of rows (median +$5.3K)	Copilot / business
Are `number_of_cus`, complexity and `rate_category` known at intake?	Decides which ablation gains are usable	Business
Does `work_order_number` rise with time?	Enables a time-based holdout	Copilot
Duplicates on the 20 inputs with differing targets	Sets the noise ceiling; may need grouped CV	Copilot
Are the $0 jobs predictable from the 20 inputs?	Zeros are mostly SEC funding decisions	Round 1 results
Top 1% of costs (up to $1.48M)	Test results with and without them	Round 1 results

Roadmap.

1. **Round 1:** run the baseline harness; report per-model and per-type KPIs.
2. **Round 2:** official KPI with the $500 rule, ablation mode, component models in the blend, optional work-order holdout.
3. **Round 3:** multi-seed CV, stacking, calibration, Optuna tuning of the top models.

4. **Round 4:** background runner with time budget and run history; single `--final` pass on the holdout.